SHIP TYPES 1.2 · ABOUT 6 MIN

Deadweight, Volume and Linear-Dimension Carriers

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Stowage Factor and Capacity Factor

A PROPERTY OF THE CARGO

SF = volume to stow 1 t

m^3/t. Ore, cement, coal, grain, oil: low. Cotton, tobacco, fruit, cars, electronics: high.

A PROPERTY OF THE SHIP

CF = hold volume / DWT

m^3/t DWT. Set by the designer, through depth, beam and the shape of the holds.

Roughly: cargo with **SF below the ship's CF** fills the deadweight first; cargo with **SF above CF** fills the holds first. The designer picks CF for the cargo the ship will carry.

What Runs Out First When You Load the Ship

Deadweight carriers

Heavy cargo, SF below about 1.3 m³/t. The load line is reached with hold space left. CF below about 1.5 m³/t DWT.

Bulk and ore carriers, crude tankers, tramp cargo ships

Volume carriers

Light cargo, SF above about 2.0 m³/t. The holds are full before the load line; excess freeboard. CF above about 2.5 m³/t DWT.

Ro-Ro, car carriers, Ro-Pax, container ships, reefers, cruise ships

Linear-dimension ships

One of L, B, T or D is fixed by a canal, a port, a river or the size of the cargo unit.

Panamax ships, Great Lakes bulk carriers, cellular container ships

Special-purpose ships

Sized by a mission: bollard pull, ice class, fish hold, deck work. No general method.

Tugs, icebreakers, fishing and offshore vessels, fast craft

Cargo between 1.3 and 2.0 m³/t, or cargoes that vary a lot: the ship can be designed either way.

When One Dimension Is Fixed from Outside

Constraint	Largest ship allowed	Fixes
Panama Canal, old locks (Panamax)	LOA 294.13 m · B 32.31 m · T 12.04 m	Beam, draught
Panama Canal, new locks (Neopanamax)	LOA 366 m · B 51.25 m · T 15.2 m	Beam, draught
St Lawrence Seaway (Seawaymax)	LOA 225.5 m · B 23.8 m · T 8.08 m	Beam, length
Azores island ports	Draught and berth length, each port	To be checked, port by port
ISO container (ISO 668)	8 ft = 2.438 m wide; 8 ft 6 in standard, 9 ft 6 in high-cube	Beam and depth, in steps

> The cost: dimensions jump in steps and the proportions suffer. A 3rd-generation Panamax container ship of about 3,700 TEU: B = 32.2 m, L = 245 m, **L/B = 7.6**.

Which Category Is the Reference Ship?

Is it the weight of the cargo or the tank volume that runs out first?

Which line of the brief fixes a main dimension from outside?

Do the ports on the route limit draught or length?

So: does the design start from deadweight or from volume?